Course Syllabus

DATA 271 Data Wrangling and Visualization

Spring 2026

Course Meeting Information

This course meets:

Lecture: MoWeFr 2:00PM - 2:50PM, BSS 302,

Activity: Tuesday 11:00AM - 12:50PM, BSS 313

Final exam: 5/11/2026, Monday, 12:40PM - 2:30PM, BSS 302

Please be sure to regularly check Canvas and your Cal Poly Humboldt email for course information and updates.

Instructor Information

Name: Emily Chang

Email: ekc140@humboldt.edu (please include "*DATA 271"* in subject when you email me about this course)

Office Hours: Tuesdays 1pm - 3pm (BSS 316)

other hours TBA :)

Course Description

Catalog description:

This course provides an intensive, hands-on introduction to data wrangling, cleaning and visualization using a contemporary programming language. You will learn the fundamental skills required to acquire, transform, manipulate, and visualize data in a computing environment that fosters reproducibility. The overall goal is to create actionable data from raw sources and then perform exploratory analysis. These steps

will include importing data, viewing data diagnostically, identifying outliers, imputing data, cleaning data, calculating basic statistics and creating informative plots.

A minimum grade of a C- is required to count towards the Data Science major.

Prerequisites: Data 111 and CS 111.

The deadline to drop without a serious and compelling reason is 11:59 p.m. on Monday of Week 2 of instruction. Refer to the Academic Calendar for the exact date.

Learning Outcomes & Competencies

Course Learning Outcomes. Upon completion of this course, you will be able to:

- Perform your data analysis in a literate programming environment with a variety of data structures.
- Import and manage structured and unstructured data and adapt data to a desirable format for analysis and visualization
- Manage different data structures
- Import, scrape, impute, and export data
- Index, subset, slice, reshape, manipulate, transform, and summarize your data
- Join disparate data sources
- Methodically explore and visualize data
- Create effective graphs and plots using "the grammar of graphics"
- Use current analysis, presentation, and collaboration tools in the Data Science field to communicate with data scientists and other parties (e.g. GitHub, Jupyter Notebooks)
- Describe where data wrangling and visualization reside within the Data Science profession

This course contributes to the achievement of Program Learning Outcomes (PLOs) and Institutional Learning Outcomes (ILOs). If applicable, this course also counts as [insert GEAR area(s)]. Review learning outcomes and competencies at the University Syllabus Addendum.

Course Readings, Materials & Fees

Textbook(s):

Both textbooks below are optional. They are helpful resources, but regular reading is not required.

- *Python Data Cleaning Cookbook* by Michael Walker and Data

This book is available for free through the library here

Links to an external site.

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- *Hands on Data Analysis with Pandas* by Stefanie Molin

This book is available for free through the library here

Links to an external site.

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Additional Materials: Readings and other materials such as assignments, labs, and slides will be provided on Canvas.

Computer and Internet: You will need a computer and internet access. You may use campus computer labs and, if needed, you may borrow laptops. Please contact me if you need a laptop.

Software needed for this course: Access to Python. Python is free, and is available on computers in our computer labs, or can be downloaded for free to your individual computer. We will be using a cloud-based JupyterHub for this class, however, installation of Anaconda is highly encouraged.

Course Topics & Schedule

The schedule below is preliminary and is subject to change with fair notice. Continue to check Canvas for an updated schedule.

Week	Day	Main Topics
1	Wed Jan 21	Data Structures
	Fri Jan 23	Github
2	Mon Jan 26	List and Dictionary Comprehension

	Wed Jan 28	Loops/Conditioning using Comprehension
	Fri Jan 30	Syntax for Regular Expression
3	Mon Feb 2	Python's re module
	Wed Feb 4	Markdown in Python and Reproducible Research
	Fri Feb 6	NumPy
4	Mon Feb 9	Applications of NumPy
	Wed Feb 11	Method Chaining and Intro to Pandas
	Fri Feb 13	Using Series and DataFrames
5	Mon Feb 16	Importing Data from CSV or Excel
	Wed Feb 18	Subsetting/Filtering Data with Pandas
	Fri Feb 20	Visualization: Matplotlib Basics
6	Mon Feb 23	Review for Midterm
	Tue Feb 24	**MIDTERM 1**
	Wed Feb 25	More on Matplotlib

	Fri Feb 27	Grammar of Graphics
7	Mon March 2	Grammar of Graphics, part 2
	Wed March 4	Wrap up Grammar of Graphics
	Fri March 6	Intro to Exploratory Data Analysis
8	Mon March 9	EDA Case Study
	Wed March 11	Advanced Plotting: Box/Violin Plots for EDA
	Fri March 13	More Advanced Plotting for EDA
9	Mon March 23	Creating New Variables
	Wed March 25	Importing Data with APIs
	Fri March 27	Creating New Variables
10	Mon March 30	Lab 9: APIs
	Tue March 31	*Campus Holiday: NO CLASS*
	Wed April 1	More on Cleaning Data
	Fri April 3	Cleaning Data: indexing, sorting, subsetting
11	Mon April 6	Review for Midterm

	Tue April 7	**Midterm 2**
	Wed April 8	Reshaping and Combining Data
	Fri April 10	More on Combining Data
12	Mon April 13	Web Scraping
	Wed April 15	Importing Data from a database with SQL
	Fri April 17	Importing Data from a database with SQL, part 2
13	Mon April 20	Peer Review of Projects
	Wed April 22	Peer Review of Projects
	Fri April 24	Interactive Plotting
14	Mon April 27	Interactive Plotting and Dashboards
	Wed April 29	Building Dashboards
	Fri May 1	Types of Missing Data & Why It Occurs
15	Mon May 4	Handling Missing Data With Pandas
	Wed May 6	Prepare for Final Presentations
	Fri May 8	Prepare for Final Presentations

16	**Final**: Mon May 11	Final Project Presentations

Course Structure & Modality

- This course includes in-person meetings on-campus. Class time will include lectures, discussions, and activities. This course will also include work on Canvas, where you will find a detailed schedule, links to course materials, assignment instructions, and so on. Each week, be sure to review Canvas before class so that you are prepared for in-class work and upcoming assignments.
- This is a 4-unit course, and thus you should expect at least 180 hours of work during the semester, which equates to an average of at least 12 hours per week between in-class/instructional activities, review of materials, and completion of assignments.

NOTE: Class meets 3 hours per week for lecture and 2 hours per week for activity, so expect at least 7 hours per week of work outside of class.

Assignments:

Attendance and Participation:

Regular attendance is **mandatory** for this course and is a vital component of your success in this course. Attending classes offers several benefits, including:

1. Active Learning: In-class engagement promotes active learning through discussions, activities, and hands-on exercises
2. Immediate Feedback: Attending class allows you to seek clarification, ask questions, and receive feedback
3. Collaborative Environment: Interacting with your peers fosters shared knowledge, diverse perspectives, and peer-to-peer learning
4. Access to Important Information: Class attendance ensures you stay informed about updates, announcements, and additional resources.
5. Building a learning community: Regular attendance helps you establish connections with your fellow students and build networks beyond the course.

Remember, attendance is not solely about fulfilling a requirement but about taking an active role in your learning journey. Your presence and participation contribute to a rich and engaging learning environment that benefits both you and your peers. We want you to be here!

Exceptions and Accommodations:

If unforeseen circumstances prevent attendance, please communicate with me *as soon as possible* to discuss necessary accommodations or alternative arrangements. I am happy to work with you.

Exams: There will be two in-class coding exams. A study guide and details for each exam will be provided in class and on Canvas. They will be held during lab sessions in Week 6 and Week 11.

Homework: Problem sets will be assigned approximately bi-weekly and collected through a file upload on Canvas. Homework assignments are due on Fridays. Your lowest score will be dropped.

Lab activities: Lab reports will be assigned and collected through a file upload on Canvas. The file upload will include both a notebook of a polished lab report as well as Python code used to generate figures, statistics, and other information in your lab report. Your lowest score will be dropped. Lab activities will be due on Thursdays.

Project: There will be 1 large project for you to work on throughout the semester. Your project idea is due in week 5, your project checkpoint is due in week 10, and your **final project is due Friday, May 8**. More information is in the project description and specific dates are on Canvas.

Notes:

- If you have problems submitting any assignments on Canvas or have other technical questions, contact Canvas 24/7 Support (help button in Canvas) or the Cal Poly Humboldt Help Desk (help@humboldt.edu or 707-826-4357).
- If you have questions about the course, such as concepts covered, your assignments, or scores, please contact me by email, during office hours, or by appointment.
- Review of your work will not be done publicly. If assignments ask you to submit work in a public space that is visible to people outside of the course, please contact me if you would like to discuss alternative options.

Grading & Evaluation:

Example language for required information:

Assignment Category	% Course Grade

Homework	25%
Lab activities	25%
Midterms (2)	10% each
Project Idea	5%
Project Checkpoint	5%
Final Project	20%

You will earn a grade in this course based on the following scale:

A = 93% or above;	**A-** = 92.9-90%	
B+ = 89.9-87%;	**B** = 86.9-83%;	**B-** = 82.9-80%
C+ = 79.9-77%;	**C** = 76.9-73%;	**C-** = 72.9-70%
D+ = 69.9-67%;	**D** = 66.9-60%;	**F** = <60%.
For CR/NC, 70% or above is needed to pass		

Grade Mode & Requirements:

Incomplete grades will not be allowed for this class. If you think you may not be able to finish the work in the course, it would be best if you dropped the course before the final withdrawal date.

General Grading Guidelines

- You will be able to track your grades using the "Grades" link in our course Canvas.
- If you have any questions about your grades, feel free to contact me.
- Content, accuracy, completeness, evidence of engagement with materials, and evidence of effort will be the primary basis for grading any subjective components of assignments.

- Assignments should be overall well-written, well-organized, and edited for typos.
- You are encouraged to explore outside information, but unless directed otherwise, assignments should be based on (and will be graded according to) course materials.
- There will be a few opportunities for extra credit throughout this course on exams and activities.

Late / Make-Up Policy

If you have an emergency, please notify me immediately if this affects your coursework or attendance. Late work will be accepted only with a serious and compelling reason and documentation. I will ask you to work with the CARE office in this situation. Their information is here: https://deanofstudents.humboldt.edu/CARE.

For help documenting or receiving services for serious, confidential, or ongoing circumstances, please contact the Dean of Students office https://deanofstudents.humboldt.edu

Policies, Procedures, and Resources

- **Positive Learning Environment:** It is essential that we create a positive learning environment in which we are all respectful of one another. You should not hesitate to ask questions nor feel embarrassed when you get the wrong answer. We all come from different backgrounds and have different learning styles. **We learn most through trying, puzzling, struggling and making mistakes**. You are a very important part of the class, and your ideas, questions, and mistakes help us all. We can help each other solve problems, stay on schedule and be motivated. Math is way more fun in a group!
    1. **Inclusivity:** Students in this class are encouraged to speak up and participate in class and on assignments outside of class. Each of us must show respect for each other because our class represents a diversity of beliefs, backgrounds, and experiences. I believe that this is what will enrich all of our experiences together. I recognize that our individual differences can deepen our understanding of one another and the world around us, rather than divide us. In this class, people of all ethnicities, genders and gender identities, religions, ages, sexual orientations, disabilities, socioeconomic backgrounds, regions, and nationalities are strongly encouraged to share their rich array of perspectives and experiences. If you feel

your differences may in some way isolate you from our classroom community or if you have a specific need, please speak with me early in the semester so that we can work together to help you become an active and engaged member of our class and community.

2. **Expectations of the Student**
3. Log into the Canvas frequently. Attend class regularly. Schedule regular time to work on assignments, read the text, and study.
4. Prepare to the best of your ability for every aspect of this course.
5. Take the opportunity to learn how to write your own thoughts and code; don't plagiarize. Be sure to give credit where credit is due and cite your sources or use footnotes or endnotes. In particular, do not post material or take solutions from Chegg or Course Hero or copy code from classmates.
6. Learning through collaboration (defined as working with or learning from another) is an effective tool used in this class and in your future employment. Collaboration on assignments is encouraged, writing individual code, solutions and lab reports is required.
7. **Expectations of the Instructor** If we all live up to our academic responsibilities, this course will be meaningful for all who participate. Please feel free to discuss these points with me at any time during the course this semester.
8. I will prepare and review course materials to be as current and accurate as possible.
9. I will be available to answer questions or issues that may arise for you during this course. Expect a 48-hour turnaround time for response to emails during the work week. I do not always read and reply to emails in the evening or on the weekends or holidays.
10. I will try to the best of my ability to prepare you for the exams and other assessments in this course.
11. I will utilize fair and honest evaluation techniques for each assignment required for this course.
12. To the best of my ability, I will make this a valid and worthwhile learning experience.
13. I will do my best to address the needs of a diverse range of learning styles in this course.

- **Emergency Procedures:** Review the evacuation plan and emergency procedures for the classroom. During an emergency, information can be found on campus conditions at 707-826-INFO or the HSU Emergency website.

- **Academic Honesty:** Students are responsible for knowing policy regarding academic honesty. For more information, visit: Academic Honesty Policy
- **AI Policy:** Generative artificial intelligence (AI) programs, such as ChatGPT, may be used in this course to learn more about a topic, organize thoughts, generate extra practice problems, generate definitions, and consult with it as a virtual tutor with appropriate citations complying with Humboldt's Student Academic Honesty Procedure. You are responsible for fact-checking statements composed by generative AI models and respecting intellectual property. Generative AI may not be used for composing responses that you are instructed to write, generating code or answers directly, or writing reports. If you are uncertain about the appropriate use of generative AI in this course, you are encouraged to contact me for clarification. Inappropriate use of generative AI programs is not permitted and will be treated as plagiarism as defined in Humboldt's Student Academic Honesty Procedure and handled in accordance with the Procedure.
- **Netiquette**: For any course component involving online interaction, university regulations regarding disruptive behavior extend to the online environment. Appropriate online behavior is expected.
- **Students with Disabilities:** Persons who wish to request disability-related accommodations should contact me immediately so I can assist you in a timely manner. If you have not yet done so, please request services with the **Campus Disability Resource Center (CDRC)** in the Learning Commons, Lower Library, 707-826-4678. https://cdrc.humboldt.edu/
- **Add/Drop Policy:** Students are responsible for knowing the University policy, procedures, and schedule for dropping or adding classes. The deadline for students to change their course schedule without penalty, known as the Add/Drop date is 11:59 p.m. on the Monday after the second week of classes of the regular semester term. After the deadline, approval to add or drop a class requires approval of a documented serious and compelling reason. See the Resolution on Decoupling Drop/Add Dates from Census (22-14/15-APC), registration help andwithdrawal process for additional information.
- **Attendance and Disruptive Behavior :**Students are responsible for knowing policy regarding attendance and disruptive behavior: Class Attendance and Disruptive Behavior.
- **University Syllabus Addendum**: Students are responsible for knowing the information about campus policies, procedures, and resources on the Syllabus Addendum website linked below. The site includes topics such as learning outcomes; registration policies; academic honesty policy; attendance and disruptive behavior policy; standards for student conduct; prevention and reporting of discrimination, harassment, and retaliation; animals on campus

policy; emergency procedures; resources for students with disabilities; learning and advising resources; counseling and psychological services; financial aid; IT Help; and more.
https://academicprograms.humboldt.edu/content/syllabus-addendum

Additional Information

Information on the syllabus is subject to change with notice. Any significant changes will be marked in the syllabus and communicated to students via email and/or an announcement on Canvas.